The Missing Parts:

Augmenting Fact Verification with Half Truth Detection

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HUMANE Lab 석사과정 고경빈

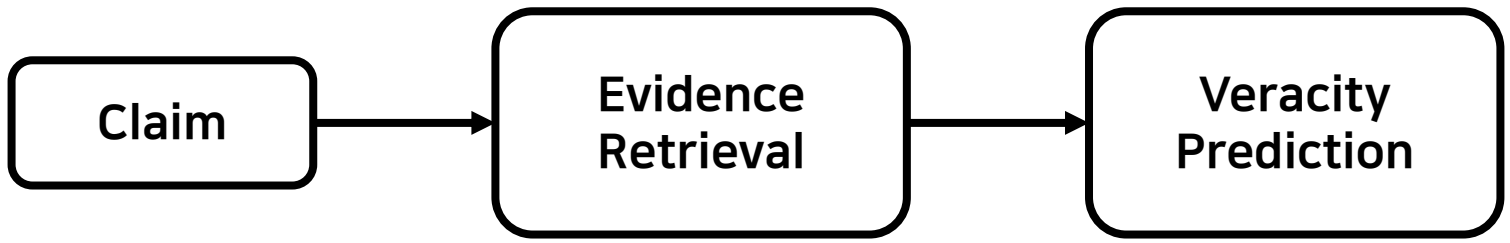
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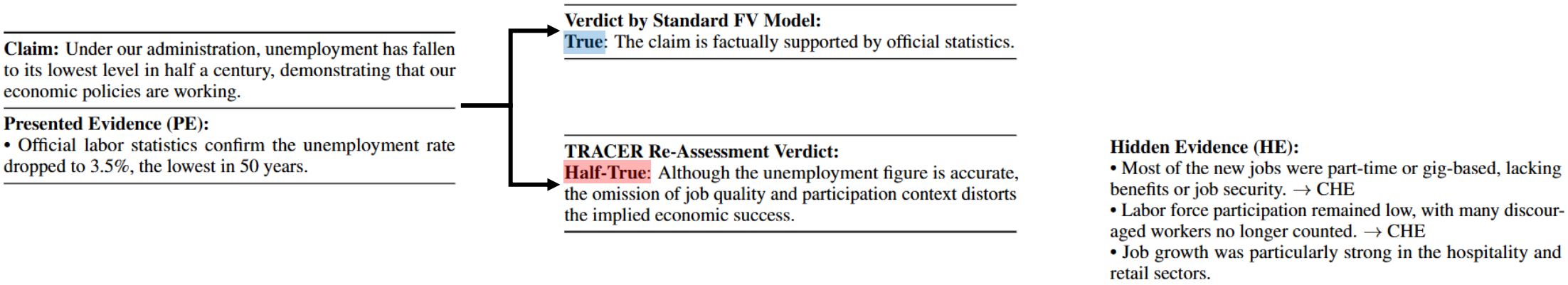
Traditional Fack-Checking Systems

- Standard Paradigm: Given a claim and a body of evidence, the Fact-Check system classifies the claim as true, false, or not enough information
- These systems are effective in identifying clearly incorrect claims



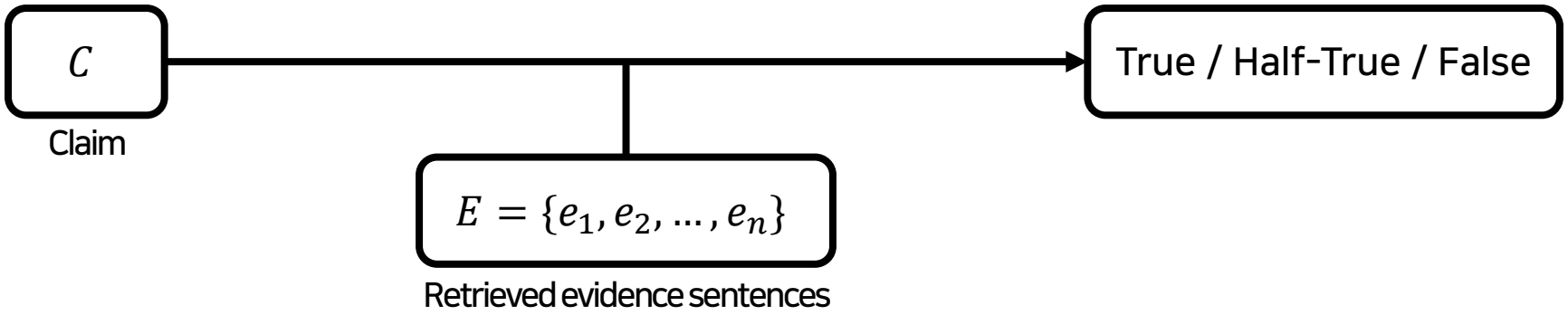
Problems of Traditional Fack-Checking Systems

- However, many real-world claims are not outright false but are still misleading due to the omission of critical context
- Traditional systems can't handle this omission-based misinformation



Half-Truth Detection

- Goal: Identify information omissions in factual claims and assess their effect on the implied message



- Classification is determined by factual support and the presence or absence of Critical Hidden Evidence
  - Presented Evidence (PE): Sentences in E explicitly stated or clearly implied in the claim
  - Hidden Evidence (HE): Sentences in E relevant to the claim but not mentioned
  - Intent: The claim's implied message conveyed to the reader
  - Critical Hidden Evidence (CHE): A subset of HE whose revelation would significantly affect the plausibility of the claim's intent

POLITIFACT-HIDDEN

- Data source: PolitiFact
  - Claim, Verdict article, Ruling, Sources
  - Rating (6): True / Mostly True / Half-Ture / Mostly False / False / Pants on Fire



Rob Sand

stated on August 24, 2026 in an ad:

Iowa once had "the best public schools in the country," but is now "in the bottom half."

2026 MIDTERMS | EDUCATION | IOWA | ROB SAND

FILE - Iowa Democratic gubernatorial candidate Rob Sand speaks during a campaign rally, June 7, 2026, in Des Moines, Iowa. (AP)

Our ruling

Sand said Iowa once had "the best public schools in the country," but is now "in the bottom half."

Several institutions rank public education by state, and by some metrics, Iowa placed highly in the past, but not number one. The state's rankings have dropped over the last 30 years; one group ranks Iowa's K-12 schools 26th nationwide.

We rate Sand's statement Half True.

Our Sources

Rob Sand, campaign ad and social media post, accessed Sept. 3, 2026

Rob Sand for Iowa, The Truer Better Agenda: Accountability for All | Rob Sand for Iowa, accessed Sept. 24, 2026

Education Data Initiative, U.S. Public Education Spending Statistics 2026, accessed Sept. 24, 2026

POLITIFACT-HIDDEN: Label Schema

- Consolidating ratings into 3 labels

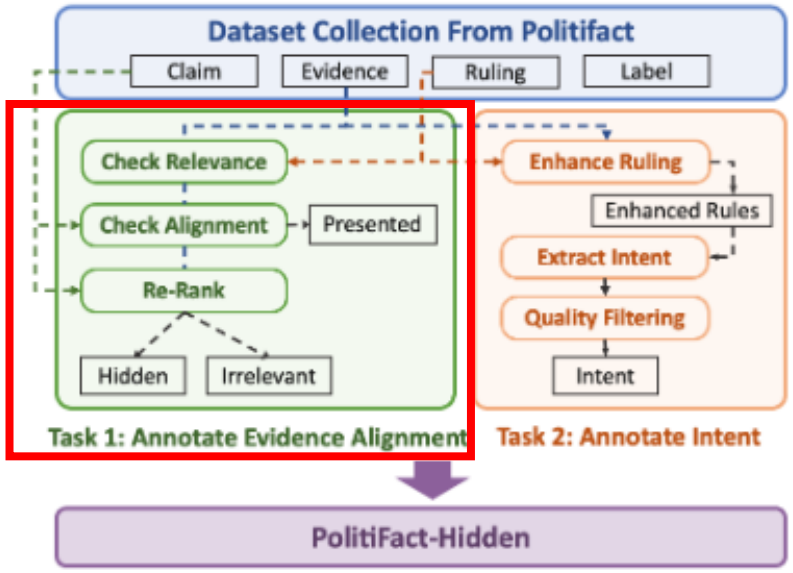
Consolidated Label	Original Rating(s)
True	True
Half-True	Mostly True, Half-True
False	Mostly False, False, Pants on Fire

- Each articles is split into ...
  - Evidence paragraph: Provide factual context
  - Ruling paragraph: Justify the final verdict

Split	True	Half-True	False	Total
Train	1,352	4,564	6,078	11,994
Dev	64	195	741	1,000
Test	93	406	1,501	2,000

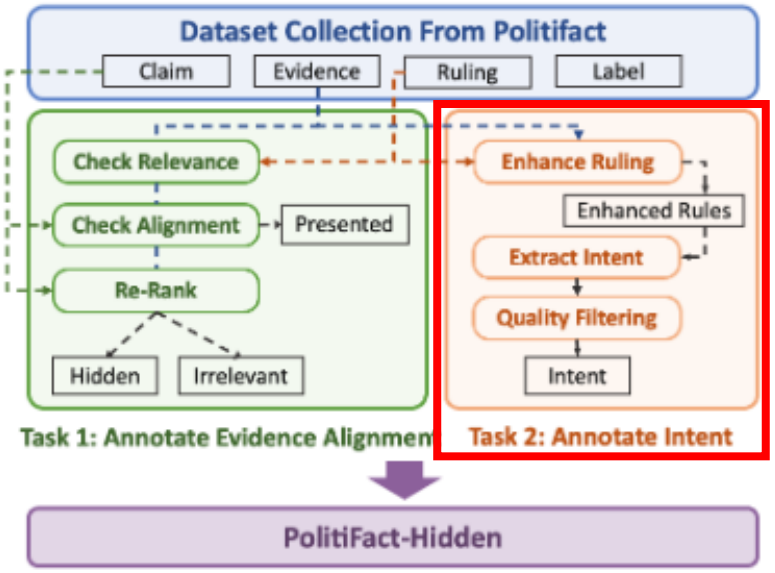
POLITIFACT-HIDDEN: Evidence Annotation

- Goal: Is evidence sentence already reflected in the claim?
- Process
  - Relevance Check
  - Presentation Check
  - Similarity Refinement

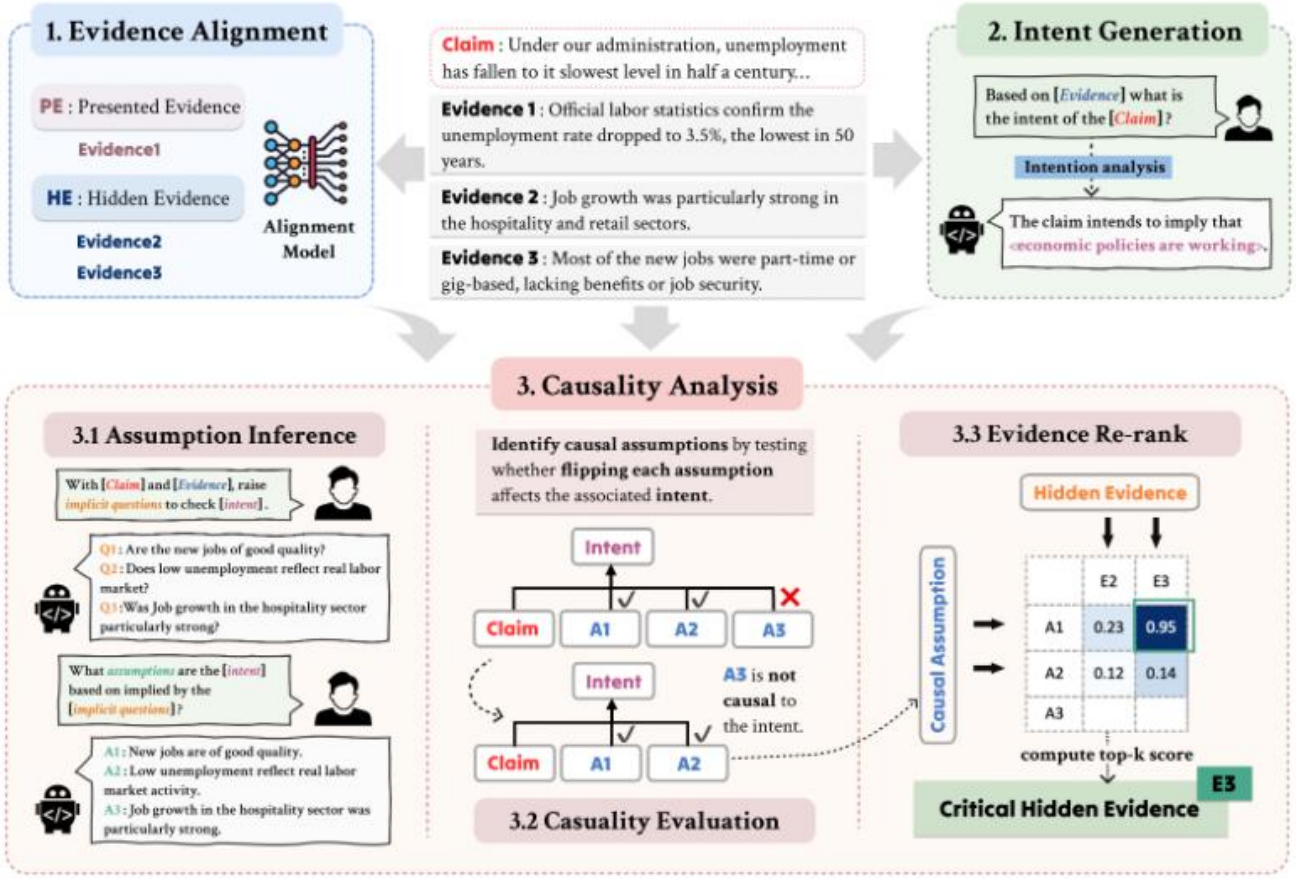


POLITIFACT-HIDDEN: Intent Annotation

- Goal: What is the claim's Intent?
- Process
  - Ruling Enhancement
  - Intent Extraction
  - Quality Filtering
    - Plausibility: The intent must not contradict the claim
    - Implicity: The intent should be implied, not overtly stated
    - Sufficiency: The intent must be specific and informative
    - Readability: The intent must be clearly and fluently expressed

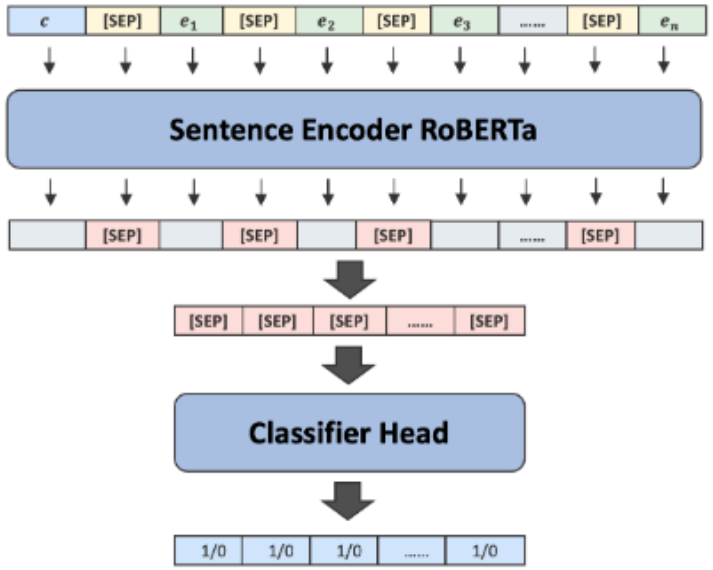


TRACER(Truth ReAssessment with Critical Hidden Evidence reasoning)



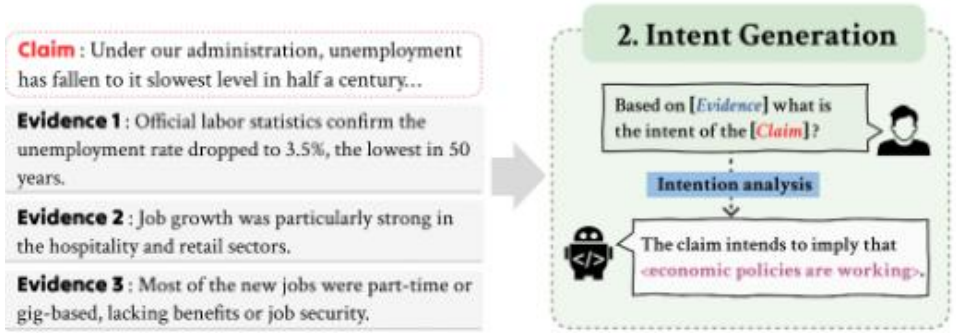
TRACER: Evidence Alignment

- Determine whether each evidence sentence is explicitly or implicitly reflected in the claim
- Fine-tuning RoBERTa-large
  - Context-aware batch sampling strategy
- Enable TRACER to isolate potentially omitted but relevant information(HE)



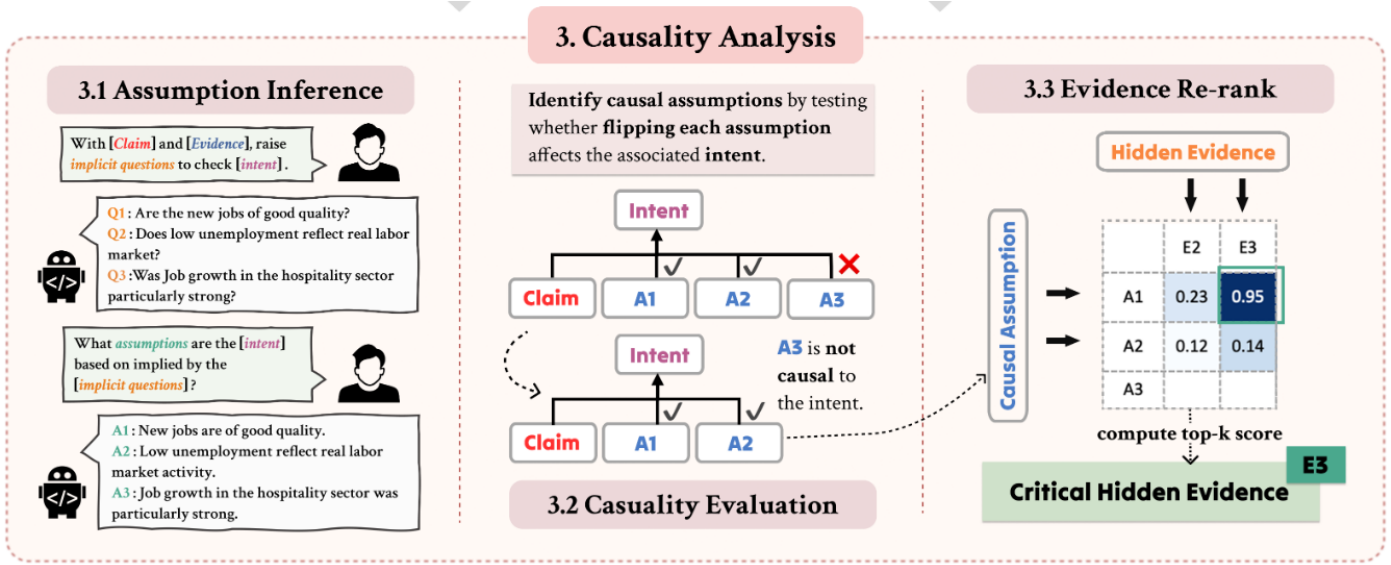
TRACER: Intent Generation

- Fine-tuning GPT-4o-mini
- Resulting intents serve as semantic anchors for next stage



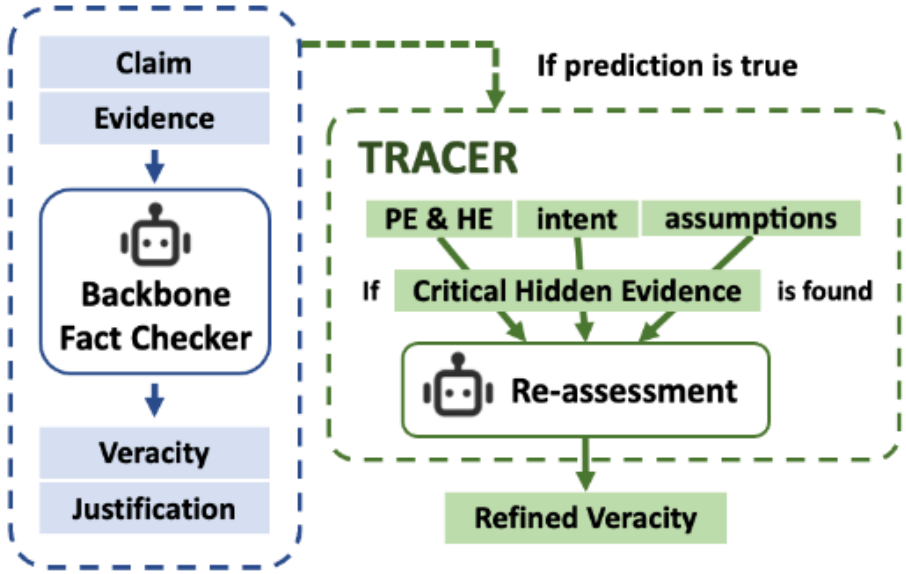
TRACER: Causality Analysis

- Estimate the causal inference of each HE sentence on the inferred intent to distinguish CHE
- Process
  - Assumption Inference: Derived from evidence through binary question and abstraction
  - Causality Evaluation: Given $do(A_i = \neg A_i)$, does the intent still hold?
  - Evidence Re-rank: Retrieve CHE based on semantic similarity and an NLI model



TRACER: Final Re-Assessment Module

- If no CHE is found, the original label is preserved
- If CHE alters the plausibility of the intent, TRACER reclassifies the claim as Half-True or False



Baselines

- Reasoning methods
  - Chain-of-Thought(CoT)
  - HiSS
- Agent-based fact-checkers
  - QACheck
  - ProgramFC
- Prompting-based LLMs
  - CHECKWHY
  - Flan-T5

Overall Performance

- TRACER improves both accuracy and macro-F1when added to reasoning-based backbones

Method	Accuracy		F1	
	Dev	Test	Dev	Test
QACheck	48.5	48.8	38.0	38.6
ProgramFC	55.4	56.9	32.9	34.2
CHECKWHY	74.8	65.9	64.2	54.6
Flan-T5	69.7	70.0	50.8	50.4
CoT	76.6	76.3	68.5	64.3
CoT +RA	77.3	78.5	**68.7**	**68.0**
Improvement	*↑ 0.7*	*↑ 2.2*	*↑ 0.2*	*↑ 3.7*
HiSS	76.3	78.3	60.3	59.4
HiSS +RA	**78.1**	**81.9**	64.3	65.7
Improvement	*↑ 1.8*	*↑ 3.6*	*↑ 4.1*	*↑ 6.3*

Half-True Detection

- TRACER substantially enhances performance on the Half-True class
- Agent-based baselines have trouble capturing hidden context

Method	Dev			Test		
	Precision	Recall	F1	Precision	Recall	F1
QACheck	24.0	55.1	33.4	24.3	54.4	33.6
ProgramFC	11.8	2.0	3.5	18.2	6.9	10.0
CHECKWHY	43.2	76.5	55.3	34.3	58.1	43.1
Flan-T5	37.7	33.7	35.6	44.1	27.3	33.7
CoT	44.6	71.8	55.0	45.0	63.8	52.8
CoT +RA	45.5 ↑ *0.9*	**83.6** ↑ *11.8*	**59.0** ↑ *4.0*	48.5 ↑ *3.5*	**79.3** ↑ *15.5*	60.2 ↑ *7.4*
HiSS	34.9	47.6	40.2	53.7	37.9	44.4
HiSS +RA	**46.3** ↑ *11.4*	54.4 ↑ *6.8*	50.0 ↑ *9.8*	**55.3** ↑ *1.6*	66.8 ↑ *28.9*	**60.5** ↑ *16.1*

Pre-Class Performance

- TRACER substantially improves the classification of Half-True claims without performance loss on other claims

Method	True	Half-True	False
CoT	52.9	52.8	**87.1**
CoT +RA	**56.7** ↑ *3.8*	**60.2** ↑ *7.4*	**87.1** *(–)*
HiSS	44.7	44.4	88.9
HiSS +RA	**46.6** ↑ *1.9*	**60.5** ↑ *16.1*	**90.1** ↑ *1.2*

Ablation Study

- ②: Intent modeling alone provides meaningful signals
- ③: Breaking down intent into assumption is helpful
- ④: Causality checking effectively suppresses noisy or irrelevant assumption

Cfg	Intent	Assump.	Causal.	F1 (H)	F1
①	–	–	–	44.4	59.4
②	✓	–	–	50.9	64.7
③	✓	✓	–	**61.2**	61.7
④	✓	✓	✓	60.5	**65.7**

Summary

- Formulate half-truth detection as a fact verification task
- Introduce POLICIFACT-HIDDEN, a new benchmark of detecting half-truth
- Propose TRACER, a framework that detects omission-based misinformation via evidence alignment, intent modeling and causal reasoning

My Opinion

- 기존 가짜 뉴스 탐지 벤치마크는 대부분 가짜 정보 유무가 핵심이었는데, 현실적으로 더 많이 사용되며 큰 위협이 될 수 있는 Half-Truth까지 다루는 벤치마크를 제안한 것이 가장 큰 기여점이라고 생각함
- 탐지하고자 하는 과업에 최적화된 PolitiFact를 활용하고, 논문에서 정의한 Half-Truth 탐지 공식에 맞게 증거와 의도를 라벨링 함. 이러한 벤치마크 구축 과정을 접한 경험이 많이 없어서 관련 벤치마크 구축 시 도움이 될 것 같음
- Half-Truth를 찾기 위해서는 의도적으로 숨긴 증거를 찾아야 하는데 의도 생성 -> 가설 추론 -> 인과성 평가 방식으로 해결한 것이 인상 깊었고, 프레임워크가 모듈식이라는 것이 큰 장점이라고 생각함
- 주요 모델이 GPT-4o-mini로 비교적 옛날 모델이고, 부록에 LLaMA2-7B 실험이 있긴 하지만 이걸로 일반화 성능을 주장하는 것이 타당한가에 대한 의문이 듦

Open Question

- Generative Montage, Half-Truth 말고 놓치고 있는 Lying with Truths가 있을까? 있다면 어떻게 해결할 수 있을까?